8(a)(i)	1. Solar energy can be affected by cloud cover, OR is only present in the day. 2. Wind speeds can vary for wind power 3. It is difficult to efficiently store excess energy generated for use at a later timing when the supply of energy falls. (any 2 of the above)	[2]
(ii)	γ-radiation is high energy/high frequency/short wavelength electromagnetic radiation / photons	[1] [1]
(iii)	γ-radiation can penetrate the skin/go through human/has high penetrating power ionizing powers can cause damage to DNA / cells / organs /tissue or cause cancer or death	[1] [1]
(iv)	To ensure that all γ-radiation would travel the same distance *x* through the absorber.	[1]
(v)	The curve reaches an asymptote at the *x*-axis. OR The (ratio of the) count rate does not go to 0.	[1]
(b)(i)	$C_X / C_0 = e^{-\mu x}$, Taking ln on both sides, $\ln (C_X / C_0) = -\mu x$	 [1] [1]

	As Fig. 8.3 is a graph of ln (C_X / C_0) against x with a straight line/constant gradient, passing through the origin, it indicates a relationship $C_X / C_0 = e^{-\mu x}$.	[1]
(ii)	gradient = $-\mu$ gradient = -4.5 / 10 = -0.45 Hence, μ = 0.45 cm^{-1}	 [1] [1]
(c)(i)	units of μ_m = units of μ / units of ρ = cm^{-1} / g cm^{-3} = g^{-1} cm^2	 [1]
(ii)	For lead, μ = 0.45 cm^{-1} $\mu_m = \mu/\rho$ = 0.45 / 11.3 = 0.0398 = 0.040 cm^2 g^{-1}	 [1]
(d)(i)	average μ_m = (0.035 + 0.037 + 0.040) /3 = 0.037 cm^2 g^{-1} For concrete, $\mu = \mu_m \rho$ = 0.037 × 2.4 = 0.037 × 2.4 × 10^3 × 10^3 / 100^3 = 0.0888 = 0.09 cm^{-1}	[1] [1] [1] [0]
(ii)	$C_X / C_0 = e^{-\mu x}$, For same shielding effect, value of C_X / C_0 is the same. Hence, value of μx must be the same. $(\mu x)_{concrete} = (\mu x)_{lead}$ (0.09) x = (0.45) (4.0) x = 20 cm OR From Fig. 8.2, when x = 4.0 cm, C_X / C_0 = 0.16 Using $C_X / C_0 = e^{-\mu x}$, ln (C_X / C_0) = $-\mu x$ ln 0.16 = -0.09 x x = 20 cm	 [1] [1] [1] [1]
(iii)	1. Concrete is cheaper OR more available than lead. 2. Concrete is a stronger material than lead 3. Concrete can be easily used in building compared to lead as it can mould into various shapes and forms 4. Lead is toxic compared to concrete. (any 2 of the above)	[2]

End of solutions